

# RISLAN MOHAMED

Associate Software Engineer · AI/ML Developer · Cybersecurity Researcher

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## PROFESSIONAL SUMMARY

Results-driven Associate Software Engineer with hands-on experience in full-stack web development, AI/ML systems, database management, and cybersecurity. Skilled in building intelligent applications using TensorFlow, Claude AI, and Python, with a strong foundation in penetration testing and network infrastructure. Completed my Pearson BTEC Higher National Diploma in Computing (Software Engineering), combining academic research with real-world industry practice. Passionate about leveraging artificial intelligence to solve complex security and operational challenges, Currently pursuing my BSc (Hons) Computer Science in Artificial Intelligence and Data Science (Top-Up Degree) .

## TECHNICAL SKILLS

|                       |                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------|
| Programming Languages | Python · Java · PHP · JavaScript · HTML5 · CSS3                                       |
| Web Development       | Bootstrap · Responsive Design · PHP Backend · REST APIs · Cross-browser Compatibility |
| AI / Machine Learning | TensorFlow · Deep Learning · Streamlit · Scikit-Learn · NumPy · Pandas                |
| Data Science & Viz    | Matplotlib · Seaborn · Plotly · Jupyter Notebook · Statistical Analysis               |
| Database Management   | MySQL · phpMyAdmin · SQL · Relational Database Design & Optimization                  |
| Cybersecurity         | Penetration Testing · Kali Linux · Metasploit · CVE Analysis · VLANs · Subnetting     |
| Tools & DevOps        | Git · GitHub · VS Code · Claude Code · Model Context Protocol (MCP)                   |
| Human Languages       | English (Fluent) · Tamil (Native) · Sinhala (Conversational)                          |

## PROFESSIONAL EXPERIENCE

Associate Software Engineer November 2025 – May 2026  
UNIX-NET-TECHNOLOGIES

- Build responsive web interfaces using HTML5, CSS3, Bootstrap, and JavaScript.
- Develop secure PHP backend solutions and manage MySQL databases.
- Handle web hosting, domain management, DNS configuration, and performance optimization.

Website Manager May 2025 – December 2025  
EuroFresh Finland (Online)

- Manage e-commerce platform — product listings, pricing, and inventory updates.
- Maintain database accuracy to ensure reliable product data for customers.

Freelance Software Engineer March 2025 – 2026  
Independent Consultant

- Deliver custom software, SQL databases, and network infrastructure for clients.
- Design VLANs, subnetting, and security configurations; produce technical documentation.

IT Administrator & Analyzer 2023 – 2025  
Afran Travels & Tours

- Maintained systems and network infrastructure ensuring 99%+ uptime.
- Enforced cybersecurity policies, managed user accounts, and provided technical support.

## RESEARCH & PROJECTS

AI-Powered Automated Penetration Testing System Nov 2025 – Jun 2026  
Technologies: Python · Claude AI · MCP · Kali Linux · CVE RAG · DeepSeek-V3.2

- Built an AI-assisted penetration testing pipeline integrating Claude AI, MCP, and Kali Linux to automate vulnerability scanning, enrichment, and analysis.

- Developed a local CVE RAG engine enriching scan outputs with CVSS scores, severity ratings, and remediation recommendations.
- Evaluated using Precision, Recall, F1 Score, and Confusion Matrix; validated with DeepSeek-V3.2. (Final Thesis – BTEC HND, BCAS Campus)
- Developed a business intelligence platform capable of processing large-scale transportation and e-commerce datasets using memory-efficient techniques.
- Implemented analytics modules for revenue tracking, customer segmentation, demand forecasting, and logistics performance evaluation through interactive dashboards.
- Applied object-oriented design principles, design patterns, and automated testing frameworks to enhance system scalability, maintainability, and reliability.

### TransitFlow Analytics System

Apr 2026 – Jun 2026

*Technologies: Java 21 · Spring Boot · Python · Streamlit · H2 Database · REST API · Pandas · Plotly · JUnit 5 · PyTest · Selenium · MVC · SOLID Principles*

- Developed a business intelligence platform capable of processing large-scale transportation and e-commerce datasets using memory-efficient techniques.
- Implemented analytics modules for revenue tracking, customer segmentation, demand forecasting, and logistics performance evaluation through interactive dashboards.
- Applied object-oriented design principles, design patterns, and automated testing frameworks to enhance system scalability, maintainability, and reliability.

### AI-Powered Fabric Type Identifier for E-Commerce

April – July 2025

*Technologies: Python · TensorFlow · Streamlit · Scikit-Learn · NumPy · Pandas · Matplotlib*

- Developed a TensorFlow deep learning model to detect cotton fabric type and estimate cotton percentage with 95.7% accuracy.
- Integrated into a Streamlit app with real-time verification, confidence visualizations, downloadable reports, and watermarked outputs.
- Reduced customer return rates by providing AI-backed product verification on apparel e-commerce platforms.

## EDUCATION

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### BSc (Hons) Computer Science in Artificial Intelligence and Data Science (Top-Up Degree)

September 2026 – Present

*BCAS Campus, affiliated with Goldsmiths, University of London*

- Programme focuses on machine learning, predictive analytics, big data technologies, data mining, and artificial intelligence, developing intelligent systems that support business decision-making and innovation.

### Pearson BTEC Higher National Diploma – Computing (Software Engineering)

Nov 2024 – July 2026

*BCAS Campus (Pearson College London)*

- Level 5 qualification covering software development, database systems, AI/ML, and cybersecurity. Served as Batch Representative and active member of BCAS IT Society (BITS).

### GCE Advanced Level – Bio Science Stream

2020 – 2023

*Addalaichenai MMV National School*

- Completed A/Ls in Biology, Chemistry, and Physics; developed strong analytical and scientific reasoning skills.

## CERTIFICATIONS

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- Anthropic – Claude Code in Action (February 2026) (Credential ID: 45g2nmewtct9)
- UniAthena – Diploma in Data Visualization Using Python (Score: 89%)
- ADBI E-Learning – Cybersecurity: Principles and Data Security Tools
- Simplilearn – Introduction to AI · Cyber Security 101 · Digital Security Fundamentals · UI/UX Basics
- Additional – Cloud Computing · Machine Learning · Web Development

## VOLUNTEERING & LEADERSHIP

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- Batch Representative – BCAS IT Society (BITS), 2025–2026: Coordinated between students and faculty; helped organise tech events, workshops, and knowledge-sharing sessions.